22 A bonus of £2100 is shared by 10 people who work for a company.
40% of the bonus is shared equally between 3 managers.
The rest of the bonus is shared equally between 7 salesmen.

One of the salesmen says,

"If the bonus is shared equally between all 10 people I will get 25% more money."

Is the salesman correct?
You must show how you get your answer.

Salesmen share 100% – 40% = 60%.

1 mark
Find the remaining percentage

Pct means percentage; GBP means pounds.

	Pct	GBP	
	100	2100	
÷10	10	210	÷10
×6	60	1260	×6

1 mark
Find 60% of £2100

	People	GBP	
	7	1260	
÷7	1	180	÷7

1 mark
Share the £1260 between 7

Compare an equal share with a 25% rise from £180.

	People	GBP	
	10	2100	
÷10	1	210	÷10

	Pct	GBP	
	100	180	
÷4	25	45	÷4

A 25% rise gives £180 + £45 = £225.

1 mark
Calculate the claimed increase

An equal share is only £210, which is £30 more.

No: £210 is less than £225, so the rise is not 25%.

1 mark
Reject the claim using the correct comparison

(Total for Question 22 is 5 marks)